

SMUG of Montreal
 20 - 159 th. Ave.
 N.D. Ile Perrot, Que.
 Canada J7V 5V6

Telephone (514) 453-0477

Vol.:3 No.:7

MAY / MAI 1987

WIRE

PIN

1
2
3
4
5
6
7

1
15
3
17
5
19
7

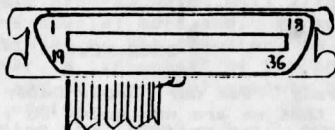


WIRE

PIN

8
9
10
11
12
13
14

21
9
23
11
25
13
14



SUPER DUPER SPECIAL

If you sell a friend a SMUG membership, you can extend your own membership for 1 year at half the cost of your regular membership fee. That's a 50% savings on your membership renewal.

Why not give it a try? You must know at least one person who has one of the following computers SVI, MSX, IBM (or compatible) or a TI-99/4A.

To take advantage of this "SUPER DUPER SPECIAL" (offered only through September 30, 1987) have your friend fill out the membership application form at the back of the newsletter (and include his/her cheque or money order) tear the form off the newsletter (with your mailing label on back still intact) and include your payment less 50% off your regular fee and send it in.

Think about it! A 50% sales commission isn't bad for selling just one membership and that's what your effort is worth to you! We may never have another offer like this again, so take advantage of it while you can!

NEWSLETTER ARTICLES

Members are encouraged to contribute to the Newsletter in the form of articles, mini programs, helpful tips, jokes, cartoons and questions. A short article may be submitted in any form in English or French. We welcome the reprinting of any article appearing in this newsletter providing credit is given to the author and to SMUG of Montreal. Opinions expressed in this newsletter are those of the writers and are not necessarily those of SMUG of Montreal. SMUG of Montreal cannot assume liability for errors in articles, programs or advertisements.

SMUG NEWS

If you've already read our "Super Duper Special" offer, then you already have an idea that we are now going to include other computer users in our group like IBMs and TI-99/4As. The reason for this is quite simple! it's because we must increase our membership if we are going to survive as a users group. The reason we have chosen IBM is because it's BASIC is very similar to that of the SVI & MSX BASICS and it's MS-DOS is a close cousin of the MSX-DOS. We have also decided to support the long lost TI-99/4A users as well, since we have a fully equipped TI sitting here doing nothing and we have well over 1,000 programs for it. So if you know of anyone with either a TI-99/4A or an IBM (or compatible), who's looking for a group to join you can recommend SMUG of Montreal to them. WE STILL INTEND TO PROVIDE THE MAJORITY OF NEWSLETTER SPACE AND SUPPORT TO THE SVI AND MSX USERS. However we need more users to keep the group going. If you have a friend who would like to see our newsletter first before joining, just send us their name and address (along with your name) and we'll be glad to send them a free sample copy of the newsletter. If they decide to join before the 30th of September 1987, we'll also allow you to extend your membership for 1 year at 50% of your normal membership dues.

Now for some good news! We have recieved a copy of Double-Sided Double-Density SVI Disk BASIC for use with 605B expanders or those who have the modified DS/DD drives & controllers, from Brian Dorofaeff in New Zealand. It is still fully compatible with the SS/DD format as well. You can order it by sending \$28.00 New Zealand currency to Brian Dorofaeff at: 42 Mahoe St., Melville, Hamilton, New Zealand

LETTERS TO THE EDITOR

Dear Sirs

My question is: I do not have an 80 column card and my CP/M will not run properly without it. Is there a modification that can be made to the bios so that the ASM, ED, DDT and CCP can be used in the 40 column mode on my SVI-328? I have tried several different times to locate this card with no success.

Sincerely Joe Alt

Dear Mr. Alt

Perhaps I can help you to solve the problem of locating an 80 column card by suggesting that you contact Mr. Gruenberg of 'Synoda Company' (see ad further on in the newsletter) for details about obtaining one or maybe you might be able to purchase one from G.S.F. HORIZONS (see last month's newsletter for address).

As for modifying the bios for this purpose, well we haven't tried to do anything of this sort, as we have an 80 column card and 99% of the CP/M programs available run in the 80 column mode. However, if there is anyone out there who has a solution to this problem we'd sure like to hear from you, as Mr. Alt isn't the only one with this problem (I'm sure), as now-a-days 80 column cards are few and far in-between.

Frances Fontaine

MSX MOUSE MOVER

Reprinted from: GREENSTAR NATIONAL COMPUTER CLUB'S NEWSLETTER
FROM: SOUTH AFRICA

Here is a little BASIC program which will allow you to make use of the mouse without the use of special software. Be sure that you depress the <LEFT> button whilst switching on the computer, so as to ensure that during the ROM check, the computer is 'informed' of the presence of the mouse.

You can of course, add any such refinements as desired, to make this little program much more useful.

Lines 41-48 determine the "jump", i.e. how many pixels moved per division and also how fast the sprite will move.

```
10 SCREEN 1:DEFINT A-Z
20 SPRITE$(0)=STRING$(8,255)
25 X=100: Y=100
30 PUT SPRITE 0,(X,Y),15,0: Z=STICK(1): IF Z=0 THEN 30
40 ON Z GOTO 41,42,43,44,45,46,47,48
41 Y=Y-1: GOTO 50
42 X=X+1: Y=Y-1: GOTO 50
43 X=X+1: GOTO 50
44 X=X+1: Y=Y+1: GOTO 50
45 Y=Y+1: GOTO 50
46 X=X-1: Y=Y+1: GOTO 50
47 X=X-1: GOTO 50
48 X=X-1: Y=Y-1: GOTO 50
50 GOTO 30
```

THE PERFECT MARRIAGE - YOUR 328 AND YOUR VCR

Reprinted from: SUGGESTIONS NEWSLETTER
FROM: U.S.A.

Statistics suggest that seven out of every ten computer owners also own a video cassette recorder (VCR). If you are among that number, then this article is for you!

There are several ways you can use your 328 with a VCR. Some of them we have taken from an article in a well known computer magazine, and some of them we have come up with on our own.

First of all, if you have your 328 connected to a color monitor, (as opposed to a TV set), you can make your monitor double as a fantastic color TV set for just a few dollars and about an hour's work! All you need is an RCA cable (the kind that hooks into any standard stereo setup) and some inexpensive two conductor speaker wire. Light duty wire will do fine, but be sure to get the stranded kind.

First, determine how many feet of speaker wire you will need to reach from your VCR to your computer monitor. In our case, it took over 50 feet of wire, and we ran the wire under the house. Be sure to allow extra for tucking the wire neatly out of view and going around corners and over doors if you don't go under the house or in the attic. Whatever figure you come up with, double it because you must run two sets of speaker wire. Next, purchase a short RCA male type patch cord (available at Radio Shack) and cut it in half at the middle. Now splice two wires coming from one of the RCA plugs to one of the 2-conductor speaker wires. (It's best to use solder, but twisting and covering with tape will work). Do the same with another of the RCA plugs and the other speaker wire. Now go to the other end of the speaker wires and repeat the process with the remaining two RCA plugs. Plug two of the RCA plugs into the audio out and video out jacks on the back of your VCR. Note which wire is plugged to audio and which is plugged to video. Next, run the wires neatly from your VCR to your monitor. Unhook your computer from the monitor and plug the remaining two RCA plugs into the proper audio in and video in jacks on your monitor.

That's it! If you have done everything properly, when you power up your VCR and monitor, you should have a great color picture with sound at your monitor! The only drawback is that you can't change channels at the monitor. If you don't want to keep changing plugs to switch between watching TV and using your computer, try using two 'Y' adapters (Radio Shack #42-2436) at the monitor. If your VCR has a TV/VCR switch, you can watch one channel on your monitor while others are watching another channel on the VCR's host TV. Remember, a lot of TV these days is unfit for anyone to watch! May we suggest a good christian channel? You can even monitor your children's TV viewing from your computer room!

There are other uses for your VCR. Plug it between your 328 and your TV/monitor (see your 328 and VCR hookup instructions) and try the following:

- 1) Record graphics or any other information straight from your computer onto your VCR. If you want to share your computer graphics with a friend who doesn't have a 328 but does have a VCR, tape your graphics and send the tape.

- 2) Try recording while you play your favourite video game. Then use your VCR's slow motion and stop features to analyze the action. This should help you improve your game strategy. If a friend is considering buying a video game, send him a sample on video tape to help him decide (it's better then pirating the program - editor's note).

3) If you make home videos with a camera, use your 328 to create lead in titles, credits, etc... This will give your videos that "professional" flavour.

4) Create a tape catalog at the beginning of your VCR tapes. Leave a minute or so of blank space at the beginning of the tape, then use your word processor, or your own computer program to make the catalog. Include the names of the shows, their length, and counter numbers to indicate where they begin and end. You might even want to include your own mini-review of a movie!

USE OF THE SPECTRAVIDEO RS232 INTERFACE TO CONTROL A SERIAL PRINTER

Reprinted from: SVI NEWS & VIEWS - By: Bernard L. Burke
FROM: South Africa

The SV 805 is a standard RS232 interface which can be used to connect your computer to a SERIAL PRINTER. The SV 805 can also be used to transfer files between computers directly or via the telephone system through a MODEM.

The function of an RS232 interface is to control the DATA FORMAT, and the DATA TRANSMISSION RATE.

With SERIAL DATA TRANSFER the computer sends out batches of BITS down a single data line (NB: with PARALLEL transfer there are 8 data lines one for each BIT of a character). Each batch of bits represents a single character.

A batch of bits consists of the following: 1) A START BIT; 2) 5 to 8 DATA bits; 3) A PARITY bit (optional); 4) 1 or 2 STOP bits.

1) The START bit is automatically generated by the RS232 and so the user need not worry about it.

2) The user must select the required of DATA bits - also known as WORD LENGTH. 7 data bits are sufficient to transmit all the ASCII characters, but if you want to use any special characters you will need 8 bits.

3) The parity bit is an optional bit which may be added as a means of error checking. It works like this: a) EVEN parity - the parity bit is set such that there will be an even number of bits (data word + parity) set to "1". The receiving device then checks the number of "1" bits to detect any errors. b) ODD parity - this is similar to even parity except that an ODD number of "1" bits are transmitted and checked. c) NO parity - the parity bit is disabled. d) MARK parity - the parity bit is always set to "1". e) SPACE parity - the parity bit is always reset to "0".

4) The STOP bits tell the receiver that a complete DATA WORD (character) has been received.

The BAUD rate is the rate at which data is transmitted. BAUD rate is measured in BITS PER SECOND (BPS).

The SV 805 has 2 programmable registers for setting up the desired conditions - these are the LINE CONTROL REGISTER and the INTERRUPT ENABLE REGISTER. In addition there is a BAUD RATE GENERATOR which must be programmed.

Proceed as follows:

- 1) Decide on the baud rate required (generally dictated by the receiving device);
- 2) Calculate the value D using the formula: $D = 192000 / \text{BAUD RATE}$;
- 3) Divide D into a HIGH BYTE (DH) and a LOW BYTE (DL) where: $DH = \text{INT}(D/256)$ $DL = D - (DH * 256)$;

- 4) Decide on the WORD LENGTH required - WORD LENGTH can be 5,6,7 or 8 bits. Calculate WL where: $WL4 = \text{WORD LENGTH} - 5$;
- 5) Select the number of STOP BITS required - one or two STOP BITS are allowed. Calculate SB where: $SB = 4 * (\text{NUMBER OF STOP BITS} - 1)$;
- 6) Decide on which PARITY mode is required - select PB such that:

NO PARITY	PB=0
ODD PARITY	PB=8
EVEN PARITY	PB=24
MARK PARITY	PB=40
SPACE PARITY	PB=56
- 7) Now calculate $CL = WL + SB + PB$; 8) The Z80A interrupts cannot be controlled from basic so the SV 805 interrupt register should be disabled -IR = 0; 9) Having set all the variables we are now ready to program the SV 805. Enter the following as a program line:


```
100 OUT &H2B,&H80 : OUT &H29,DH : OUT &H28,DL : OUT &H2B,CL :
      OUT &H29,IR
```

When entering the line you should replace the variables DH, DL, CL and IR with their respective values.

When this line is executed your SV 805 is initialised according to the chosen values.

IMPORTANT: THE SENDING DEVICE AND THE RECEIVING DEVICE MUST BOTH BE INITIALISED TO THE SAME CONFIGURATION - FAILURE TO DO THIS WILL CAUSE THE DATA TO BECOME GARBLED DURING TRANSFER.

Data transmission is through OUTPUT PORT &H28 (eg. OUT &H28,X) - all data must be in ASCII code.

IMPORTANT POINTS TO NOTE

a) Ensure that the previous character has been read by the printer before sending the next character - this is done by reading bit 5 of the MODEM STATUS REGISTER. If bit 5 is "0" then wait until bit 5 is "1". The modem status register is read from INPUT PORT &H2E - use the following code to read bit 5:

```
1000 IF (INP(&H2E)AND 32)=0 THEN 1000
```

This program line will LOOP until the printer is ready.

b) Ensure that the TRANSMITTER HOLDING REGISTER is empty before sending the next character - this is done by reading bit 5 of the LINE STATUS REGISTER. If bit 5 is "0" then wait until bit 5 is "1". The line status register is read from INPUT PORT &H2D - use the following code to read bit 5:

```
1010 IF (INP(&H2D)AND 32)=0 THEN 1010
```

This program line will loop until the transmitter holding register is empty.

LIST 1 is a simple RS232 PRINTER program for your SPECTRAVIDEO. The program illustrates the initialisation of the SV 805 and the transfer of TEXT to the serial printer. The program uses a BAUD RATE of 9600, WORD LENGTH 8 bits, NO PARITY bit, and 2 STOP bits. If your printer requires different values then change the values in LINE 100.

LIST 1

```
10 CLS: INPUT "TYPE IN TEXT TO BE PRINTED";A$
100 OUT &H2B,&H80: OUT &H29,0: OUT &H28,20: OUT &H2B,7: OUT &H29,0
900 A$=A$+CHR$(13)+CHR$(10)
910 FOR Z=1 TO LEN(A$): X=ASC(MID$(A$,Z,1))
1000 IF (INP(&H2E)AND 32)=0 THEN 1000
1010 IF (INP(&H2D)AND 32)=0 THEN 1010
1020 OUT &H28,X:NEXT:GOTO 10
```


RS232 cable configurations differ according to the devices to be connected. An IMPORTANT consideration is that the receiving device must be able to send control signals back to the transmitter. The SV 805 expects such signals on pin 6 but the printer may transmit the signals from pin 20, pin 11 or possibly another pin. Normal data is TRANSMITTED from pin 2 and RECEIVED on pin 3.

The following cable configuration should be suitable for most printers - use SUB-MIN D-TYPE 25-PIN PLUGS.

SV 805 INTERFACE

PIN 7
PIN 20
PIN 6
PIN 4
PIN 8
PIN 3
PIN 2
PIN 1

PRINTER INTERFACE

PIN 7
PIN 6
PIN 11 AND PIN 20
PIN 8
PIN 4
PIN 2
PIN 3
PIN 1

IN ADDITION YOU SHOULD CONNECT TOGETHER PINS 4 and 5 AT THE SV 805 END and AT THE PRINTER END.

This cable is known as a MODEM ELIMINATOR and is only suitable for DIRECT connection of two RS232 devices. When a MODEM is used a different cable configuration will be required.

MSX PRINTERS AND CABLES

REPRINTED FROM: MSX USER GROUP'S NEWSLETTER
FROM: ENGLAND

A reminder that any printer with a parallel interface will work with the MSX micro. Only an MSX printer, however, can print the special MSX graphics characters. These are printed out as spaces on a non-MSX printer.

MSX printer cables may not be readily available in shops but can be ordered through a computer dealer.

If you would like to make up your own lead you will need a length of 14 strand ribbon cable, a 14 pin plug for the computer and a standard 32-pin centronics plug for the printer. For connection diagram see diagram on front cover.

ONE LINERS

Reprinted from: RESEAU MSX MAGAZINE
FROM: BELGIUM

```
100 COLORC,15,C:SCREEN2:PI=3.14159:A=RND(-TIME):FORZ=0TO360STEP20:X=128+30*
COS(PI*Z/180):Y=96+30*SIN(PI*Z/180):LINE(X,Y)-(128,191):LINEX+5,Y+5)-(0,96)
:LINE(X,Y)-(128,0):LINE(X+5,Y+5)-(255,96):NEXTZ:C=(C+1)MOD14:GOTO100
```

```
100 COLOR15,1,1:SCREEN2:FORI=1TO100:A=INT(RND(1)*256):B=INT(RND(1)*192):C=
INT(RND(1)*15)+1:LINE(A,B)-(A+50,B+50),C,BF:NEXTI:FORT=1TO10000!:NEXT
```

```
100 COLOR,15,15:SCREEN2:C=(C+1)MOD14:FORX=191TO0STEP-6:LINE(32,X)-(X+32,19
1),C:LINE(223,X)-(X+32,0),C:NEXT:LINE(32,0)-(223,191),C,B:PAINT(1,1),C:PAI
NT(255,1),C:GOTO100
```

ARTICLES ET NOUVELLES

Nous encourageons les membres à contribuer à cette revue sous formes d'articles, mini programmes, conseils, dessins et questions. Vous pouvez aussi nous soumettre un article court, sous n'importe quelle forme, en français et anglais. Nous permettons la copie des articles inclus dans cette revue, à condition de mentionner l'auteurs, et le club. Les opinions exprimees sont celles des auteurs, et non celles de SMUG. Qui ne peut en aucune cas en assumer la responsabilite ni celles des erreurs dans les articles, programmes et publicite.

AU COEUR DE LA MEMOIRE CENTRALE De: RESEAU MSX MAGAZINE - BELGIQUE

Ce petit utilitaire permet à l'utilisateur de localiser une suite de deux octets dans la memoire centrale. La memoire centrale occupe les adresses decimales de 0 à 65535, ou de 0000 à FFFF en hexadecimal.

L'utilisateur du programme encode la suite de deux octets qu'il desire retrouver. Ensuite, il encode les deux adresses memoire entre lesquelles il desire que la recherche s'effectue. Des lors, l'utilitaire fouille la memoire centrale entre les adresses indiquees et il donne toutes les adresses auxquelles il a rencontre la suite de deux octets recherchee. Ce petit utilitaire est tres utile pour le programmeur en assembleur qui desire connaitre les differents endroits ou sont stockees les donnees qu'il voudrait utiliser.

Les deux octets à rechercher doivent etre encodes dans l'ordre ou l'on desire les retrouver. Ils doivent etre exprimes sous forme de deux chiffres hexadecimalux. Par exemple, si on desire rechercher l'octet EF suivi de l'octet 5, il convient d'encoder EF05.

Les adresses entre lesquelles la recherche doit s'effectuer peuvent etre exprimees en decimal ou en hexadecimal precede des signes &H. L'adresse la plus basse doit etre encodee en reponse à la question "adresse de debut" et l'adresse la plus haute, en reponse à la question "adresse de fin".

```
10 CLEAR 200,&HD000
20 AD=&HD000
30 FOR I=&HD000 TO &HD01B
40 READ A$:POKE I,VAL("&H"+A$):NEXT I
50 DEFUSR=&HD004
60 DP=0
65 FI=&H7FF0
70 CLS
80 INPUT "VALEUR EN HEXA";A$
90 INPUT "ADR DEBUT ";DP
100 INPUT "ADR FIN ";FI
110 C1$=LEFT$(A$,2)
120 C2$=RIGHT$(A$,2)
130 C1=VAL("&H"+C1$)
140 C2=VAL("&H"+C2$)
150 POKE AD+2,C1
160 POKE AD+3,C2
170 D1=DP MOD 256
180 D2=DP MOD 256
190 IF D1<0 THEN D1=D1+256
200 IF D2<0 THEN D2=D2+256
210 POKE AD,D1
```



```

220 POKE AD+1,D2
230 VO=0
240 L=USR(0)
250 V1=PEEK(&HD000)
260 V2=PEEK(&HD001)
270 V=V1+V2*256-1
280 IF VO>=V THEN GOTO 400
290 VO=V
300 IF FI>0 AND V>FI THEN GOTO 400
310 PRINT HEX$(V),
320 GOTO 240
400 PRINT:PRINT "TERMINE"
420 DATA 0,0,0,0,F3,2A,0,D0,3A,2,D0,BE,28,03,23,18
430 DATA FA,3A,3,D0,23,BE,20,F0,22,0,D0,C9

```

BANK

De: SVI - BONDWELL COMPUTER CLUB - FRANCE

Une instruction speciale du BASIC vous permet d'utiliser les bank memoires supplementaires de votre ordinateur SVI; c'est l'instruction SWITCH. Il faut toutefois pour l'utiliser que votre machine soit equipee d'une carte d'extension 64Ko. Lorsque vous tapez SWITCH, le systeme commute le bank 22 a la place du bank 02. Pour revenir au bank 02, il faut a nouveau executer l'instruction SWITCH. Cela vous permet par exemple, d'utiliser 2 programmes simultanement.

Il existe une autre facon de commuter les banks, c'est d'utiliser les ports d'Entree (Port B du PSG) qui sont affectes a cette commutation.

Les bits 0 a 4 servent a cette commutation:

- Le Bit 0 commute le Bank 1 a la place du bank 01, c'est a dire, les cartouches a la place de la ROM BASIC.
- Le Bit 1 commute le Bank 2 a la place du bank 01, c'est a dire, 32 Ko de RAM a la place de la ROM BASIC (avec extension pour un SVI-318)
- Le Bit 2 commute le Bank 22 a la place du bank 02 (meme fonction avec l'instruction SWITCH).
- Le Bit 3 commute le bank 31 a la place du bank 01 (avec extension).
- Le Bit 4 commute le Bank 32 a la place du bank 02 (avec extension).

Pour commuter, il faut:

1. Selectionner le port B du PSG. OUT &H88,15
2. Ecrire la valeur choisie dans le port B. OUT &HxC, valeur

TRUCS ET ASTUCES

De: MSX - SV328 NEWS - BELGIQUE

Voici un premier truc envoye par M. Rion qui permet de presenter un repertoire (DIRECTORY) par ordre alphabetique.

```

10 'IMPRESSION ALPHABETIQUE DES PROG.
20 ON ERROR GOTO 90
30 SCREEN0: WIDTH 38:AS="ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789"
40 PRINT "DSKF(0)":DSKF(0)
50 FOR A=1 TO 36
60 BS=MID$(AS,A,1)+"????????.???"
70 LFILES BS
80 NEXT A
90 IF ERR=53 THEN RESUME NEXT ELSE END

```

Pour sauver le contenu de votre ecran sur disque il suffit d'ecrire:
 MSX: BSAVE "ECRAN",0,&H3FFF,S SV328: SAVE "1:ECRAN",S
 Pour le recharger il suffit d'ecrire:
 MSX: BLOAD "ECRAN",S SV328: LOAD "1:ECRAN"
 Pour vider le tampon d'entree de caractere avant l'execution d'une
 commande d'INPUT, il suffit d'ecrire:
 MSX: DEFUSR=&H156: L=USR(0) SV328: DEFUSR=&H351F:L=USR(0)

TRUCS ET ASTUCES

De: GROUPE DES UTILISATEURS MSX - FRANCE

Après chargement d'un program en langage machine, depuis une cassette
 (par BLOAD,R), son adresse d'execution est indiquée par les variables
 système qui sont aux adresses 64703 et 64704 (&HFCBF et &HFCC0). On
 détermine l'adresse en commande directe par:

PRINT PEEK(64703 +256*PEEK(64704))

Connaissant cette adresse, on peut lancer le programme avec:

DEFUSR0= adresse trouvee:A=USR(0)

ADRESSES POKE INTERESSANTES POUR MSX

Pour limiter le nombre de lignes sur l'ecran a 10 par exemple:

POKE &HF3B1,10

Pour inhiber la commande STOP: POKE &HFB81,1

Pour inhiber la commande LIST: POKE &HFF87,225

RECOPIE D'ECRAN EN MSX

En SCREEN0, en langage machine pour imprimante matricielle. Ne pas
 utiliser avec un drive branche sur le micro. Lance par la touche ESC:

80 CLEAR 200, &HF000

90 A'=&HF001:POKE &HF000,F3

100 FOR N%=0 TO 62

110 READ D\$:POKE A'+N%,VAL("&H"+D\$):NEXT N%

120 DEF USR=&HF001

130 A=USR(0)

140 DATA 3E,CD,32,D1,FD,21,0E,F0,22,D2,FD,FB,C9

150 DATA FE,3A,C0,E5,C3,F5,21,00,00,01,18,28,CD

160 DATA EC,07,E5,E1,DB,98,E6,7F,FE,20,38,03,CD,A5

170 DATA 00,23,10,ED,3E,0A,CD,A5,00,3E,0D,CD,A5,00

180 DATA 06,28,0D,20,DE,F1,C1,E1,C9

190 NEW

10 REM BRONSKI BEAT:

SMALLTOWN BOY

20 REM MUSIC

30 KEYOFF

40 CLS

50 COLOR 10,1: LOCATE6,2

60 PRINT"COCORIKI BIT"

70 LOCATE6,4

80 PRINT"Interpréte:"

90 LOCATE6,7

100 PRINT"-Robot Super Sauvage

-"

110 LOCATE6,16

120 PRINT"Auteur: A.R. LONT"

130 LOCATE6,18

140 PRINT"MADE IN AMSTERDAM

150 LOCATE0,20

160 PRINT"Copyright MSX COMPUT
ER MAGAZINE-1985"

170 LOCATE0,22

180 PRINT"Publi. dans MICROS M
SX N° 5-AVRIL 86"

190 REM BRONSKI

200 SOUND7,49:SOUND6,5

210 T\$="t150R4L6406V13BV11BV7B

R64R4V13BV11BV7BR64"

220 U\$="T150L32V0B"

230 SOUND7,49:SOUND6,5

240 V\$="t150R2L3206V13BV11BV8B
v5b"

250 PLAY"XU\$:", "t125V903L1C"

260 PLAY"XUS";,"02L1A+"
 270 PLAY"XUS";,"02L1F"
 280 PLAY"XUS";,"02L1D+"
 290 PLAY"XUS";,"T13003L1C","t1
 30V9L104G"
 300 PLAY"XUS";,"02L1A+","L104F"
 310 PLAY"XUS";,"02L1F","L104C"
 320 PLAY"XUS";,"02L1D+","L104D
 +"
 330 PLAY"XUS";,"T13003L1C","T130L104G
 +","V10T130L105C"
 340 PLAY"XUS";,"02L1A+","L104F","L104
 A+ "
 350 PLAY"XUS";,"02L1F","L104C","L104F
 + "
 360 PLAY"XUS";,"02L1D+","V8L103D+","L
 104D+ "
 370 PLAY"XUS";,"T152V10L105C",
 "m2000S9T140L803C04C03C04C03C0
 4C03C04C"
 380 PLAY"XUS";,"T152V10L104A+
 +","02A+03A+02A+03A+02A+03A+02A+
 03A+ "
 390 PLAY"XUS";,"T152V10L104F",
 "03F04F03F04F03F04F03F04F"
 400 PLAY"XUS";,"T152V10L104D+
 +","03D+04D+03D+04D+03D+04D+03D
 04D "
 410 PLAY"XUS";,"M2000S9T140L80
 3C04C03C04C03C04C03C04C"
 420 PLAY"XUS";,"02A+03A+02A+03
 A+02A+03A+02A+03A+ "
 430 PLAY"XUS";,"03F04F03F04F03
 F04F03F04F"
 440 PLAY"XUS";,"T150V10M2100S9R4L8CCR
 8L8CCL16CC","M2000S9T140L803D+
 04D+03D+04D+03D+04D+03D04D"
 450 LETX=0
 460 LETX=X+1
 470 PLAY"XTS";,"M2000S9T150L80
 3C04C03C04C03C04C03C04C"
 480 PLAY"XTS";,"M2000S9T150L80
 2A+03A+02A+03A+02A+03A+02A+03A
 + "
 490 PLAY"XTS";,"M2000S9T150L80
 3F04F03F04F03F04F03F04F"
 500 PLAY"XTS";,"M2000S9T150L80
 3D+04D+03D+04D+03D+04D+03D04D"
 510 PLAY"XTS";,"M2000S9T150L80
 3C04C03C04C03C04C03C04C"
 520 PLAY"XTS";,"M2000S9T150L80
 2A+03A+02A+03A+02A+03A+02A+03A
 + "
 530 PLAY"XTS";,"M2000S9T150L80
 3F04F03F04F03F04F03F04F"
 540 PLAY"XUS";,"T150V10M2100S9R4L8CCR
 8L8CCL16CC","03D+04D+03D+04D+0
 3D+04D+03D04D"

550 PLAY"XTS";,"M2500S9T138R8L
 805CL8D+R8L8CL8D+R8D+","M2000S
 9T150L803C04C03C04C03C04C03C04
 C "
 560 PLAY"XTS";,"M2500S9T138R8L
 805CL8DR8L8CL8DR8L8D+","02A+03A
 +02A+03A+02A+03A+02A+03A+ "
 570 PLAY"XTS";,"M2500S9T138R8L
 805CL8FR8L8CL8FR8L8F+","03F04F0
 3F04F03F04F03F04F"
 580 PLAY"XTS";,"M2500S9T138R8L
 805CL8D+R8L8CL8D+R8L8D+","m200
 0S9T150L803C04C03C04C03C04C03C
 04C "
 590 PLAY"XTS";,"M2500S9T138R8L
 805CL8D+R8L8CL8D+R8L8D+","M200
 0S9T150L803C04C03C04C03C04C03C
 04C "
 600 PLAY"XTS";,"M2500S9T138R8L
 805CL8DR8L8CL8DR8L8D+","02A+03A
 +02A+03A+02A+03A+02A+03A+ "
 610 PLAY"XTS";,"M2500S9T138R8L
 805CL8FR8L8CL8FR8L8F+","03F04F0
 3F04F03F04F03F04F"
 620 PLAY"XTS";,"T150M2100S9R4L8CCR8L8
 CCL16CC","M2500S9T138R8L805CL8
 D+R8L8CL8D+R8L8D+","03D+04D+03D
 +04D+03D+04D+03D04D"
 630 PLAY"XTS";,"T150V11M2500S9L8R2C",
 "M2000S9T150L803C04C03C04C03C0
 4C03C04C"
 640 PLAY"XTS";,"T150V11M2500S9L8R2C",
 "M2000S9T150L802A+03A+02A+03A+
 02A+03A+02A+03A+ "
 650 PLAY"XTS";,"T150V11M2500S9L8R2C",
 "M2000S9T150L803F04F03F04F03F0
 4F03F04F"
 660 PLAY"XTS";,"T150V11M2500S9L8R2C",
 "M2000S9T150L803D+04D+03D+04D+
 03D+04D+03D04D"
 670 PLAY"XTS";,"T150V11M2500S9R2L8C",
 "M2000S9T150L803C04C03C04C03C0
 4C03C04C"
 680 PLAY"XTS";,"T150V11M2500S9R2L8C",
 "M2000S9T150L802A+03A+02A+03A+
 02A+03A+02A+03A+ "
 690 PLAY"XTS";,"T150V11M2500S9R2L8C",
 "M2000S9T150L803F04F03F04F03F0
 4F03F04F"
 700 PLAY"XTS";,"T150V11M2500S9R4L8CCR4
 CCL16CC","M2000S9T150L803D+04D
 +03D+04D+03D+04D+03D04D"
 710 PLAY"XTS";,"M2500S9T138R8L
 805CL8D+R8L8CL8D+R8D+","M2000S
 9T150L803C04C03C04C03C04C03C04
 C "

729 PLAY"XT\$;", "M2500S9T138R8L
 805CL8DR8L8CL8DR8L8D", "02A+03A
 +02A+03A+02A+03A+02A+03A+"
 730 PLAY"XT\$;", "M2500S9T138R8L
 805CL8FR8L8CL8FR8L8F", "03F04F0
 3F04F03F04F03F04F"
 740 PLAY"XT\$;", "M2500S9T138R8L
 805CL8D+R8L8CL8D+R8L8D+", "M2000
 05903D+04D+03D+04D+03D+04D+03D
 04D"
 750 PLAY"XT\$;", "M2500S9T138R8L
 805CL8D+R8L8CL8D+R8D+", "M2000S
 9T150L803C04C03C04C03C04C03C04
 C"
 760 PLAY"XT\$;", "M2500S9T138R8L
 805CL8DR8L8CL8DR8L8D", "02A+03A
 +02A+03A+02A+03A+02A+03A+"
 770 PLAY"XT\$;", "M2500S9T138R8L
 805CL8FR8L8CL8FR8L8F", "03F04F0
 3F04F03F04F03F04F"
 780 PLAY"XT\$;", "M2500S9T138R8L
 805CL8D+R8L8CL8D+R8L8D+", "M2000
 05903D+04D+03D+04D+03D+04D+03D
 04D"
 790 PLAY"XT\$;", "M2000S9T150L80
 3C04C03C04C03C04C03C04C", "T150
 V10L105G"
 800 PLAY"XT\$;", "02A+03A+02A+03
 A+02A+03A+02A+03A+", "L105F"
 810 PLAY"XT\$;", "03F04F03F04F03
 F04F03F04F", "L105G+"
 820 PLAY"TI50V10M2100S9R4L8CCR
 8L8CCC", "03D+04D+03D+04D+03D+0
 4D+03D04D", "L105G"
 830 PLAY"XT\$;", "M2000S9T150L80
 3C04C03C04C03C04C03C04C", "T155
 L105G"
 840 PLAY"XT\$;", "02A+03A+02A+03
 A+02A+03A+02A+03A+", "L105F"
 850 PLAY"XT\$;", "03F04F03F04F03
 F04F03F04F", "L105G+"
 860 PLAY"TI50V10M2100S9R4L8CCR
 8L8CCC", "03D+04D+03D+04D+03D+0
 4D+03D04D", "L105G"
 870 PLAY"XT\$;", "M2000S9T150L80
 3C04C03C04C03C04C03C04C", "T150
 V11R8L806C05A+06CD+FFC"
 880 PLAY"XT\$;", "02A+03A+02A+03
 A+02A+03A+02A+03A+", "L8R806C05
 A+06CD+FF"
 890 PLAY"XT\$;", "03F04F03F04F03
 F04F03F04F", "L8R806C05A+06CD+F
 FD+"
 900 PLAY"TI50M2100S9R4L8CCR4L8
 CC", "03D+04D+03D+04D+03D+04D+0
 3D04D", "L8R806F+FD+F+L16FL8D+"
 910 PLAY"XT\$;", "M2000S9T150L80
 3C04C03C04C03C04C03C04C", "T150
 V11R8L806C05A+06CD+FFC"

920 PLAY"XT\$;", "02A+03A+02A+03
 A+02A+03A+02A+03A+", "L8R806C05
 A+06CD+FF"
 930 PLAY"XT\$;", "03F04F03F04F03
 F04F03F04F", "L8R806C05A+06CD+F
 FD+"
 940 PLAY"XT\$;", "03D+04D+03D+04
 D+03D+04D+03D04D", "L8R806F+FD+
 F+L16FL8D+"
 950 IFX=1THENGOTO970
 960 GOTO460
 970 PLAY"XT\$;", "M2000S9T150L80
 3C04C03C04C03C04C03C04C", "T150
 V10M2500S9R8L805GFGA+L1606C05A
 +"
 980 PLAY"XT\$;", "03D+04D+03D+04
 D+03D+04D+03D+04D+", "R8L805GF
 G+L1606C05A+"
 990 PLAY"XT\$;", "03F04F03F04F03
 F04F03F04F", "R8L805GFGA+L1606C
 05A+"
 1000 PLAY"XT\$;", "02G+03G+02G+0
 3G+02G+03G+02A+03A+", "R8L805GF
 GA+L1606C05A+"
 1010 PLAY"XT\$;", "M2000S9T150L8
 03C04C03C04C03C04C03C04C", "T15
 0V9M2500S9R8L805GFGA+L1606C05A
 +"
 1020 PLAY"XT\$;", "03D+04D+03D+0
 4D+03D+04D+03D+04D+", "R8L805GF
 GA+L1606C05A+"
 1030 PLAY"XT\$;", "03F04F03F04F0
 3F04F03F04F", "R8L805GFGA+L1606
 C05A+"
 1040 PLAY"XT\$;", "02G+03G+02G+0
 3G+02G+03G+02A+03A+", "R8L805GF
 GA+L1606C05A+"
 1050 GOTO450

SPECTRA MSX

READ ALL ABOUT MSX COMPUTING AND
INCREASE YOUR PROGRAMMING SKILLS BY
SUBSCRIBING TO SPECTRA MSX
MONTHLY COMPUTER MAGAZINE!

PROGRAM LISTINGS

SOFTWARE REVIEWS

MACHINE CODE COLUMN

MSX HISTORY

TELECOMMUNICATIONS

CLASSIFIEDS

HARDWARE REVIEWS

Z80 ARCHITECTURE

LETTER SLOT

COMPUTER CLUB REPORT

PLUS MORE COMING...

Mail to: Spectra MSX magazine
P.O.Box 3153,
Christchurch.
New Zealand.

YES! Make me a **subscriber** to Spectra MSX magazine! Please send me a full year (12 issues) of the magazine for \$39.00 (NZ Currency).

Name _____

Address _____

Country _____

Zone _____

Phone () _____

Please make **money order** payable to: IMPAK Computing

CATERS FOR: SVI318 / SVI328 / MSX USERS...

S Y N O D A C O M P A N Y
300 CANAL BANK RD. UNIT 334
VILLE ST. PIERRE, QUE.
CANADA H8R 1H3

TELEFAX (514) 365-1610

TELEPHONE (514) 366-1195

TELEX 055-66465(SYNODA)

SUPER CLUB SPECIAL!!!

SVI - 328 64K computers -----	\$130.00
SVI - 318 32K computers -----	\$ 99.00
SVI - 601 7 Slot Expander -----	(Call)
SVI - 602 Single Slot Expander -----	\$ 19.00
SVI - 904 Mono Data Recorder -----	\$ 29.00
SVI - 903 Stereo Data Recorder -----	\$ 32.00
SVI - 105 MSX & SVI Graphic Tablets -----	\$ 99.00
SVI - 902 Single Side Double Density Drives --	\$150.00
SVI - 801 Disk Controller Card -----	\$ 35.00
SVI - 901 Printer Interface Card -----	\$ 34.72
SVI - 901 Dot Matrix Printer -----	\$180.00
SVI - 905 Half Height SS/DD Drives for 605 ---	\$135.00
SVI - 605 A or B Expanders -----	(Call)
SVI - 101 Joysticks -----	\$ 5.56
SVI Cassette Programs -----	\$ 4.95
SVI - 802 Centronic Interface Card -----	(Call)
SVI - 609 Local Area Network -----	(Call)

VALUE PAK - SV328 & 904 recorder & 10 programs \$160.00

BONDWELL 36 - IBM - XT compatible with 1 multi card, 2 DS/DD - 360k drives
640k RAM, 1 20 megabyte hard drive plus controller \$2,800.00

BONDWELL 2 - Lap-Top computer with LCD 80 x 25 char. screen, 1 3.5" built-
in disk drive, centronic interface, battery powered (Call)

SMUG of Montreal
28 - 159 th. Ave
N.D. Ile Perrot, Quebec, Canada J7V 5V6

We are the official Spectravideo and MSX user's group for the Montreal area as recognized by Spectravideo Canada, Yamaha Canada and Bondwell International. A family membership entitles you and your family to attend groups meetings, trade programs and receive 1 copy monthly of our newsletter per family. An Individual membership is for individuals who wish to attend the groups meetings, trade programs and receive a copy of our monthly newsletter. Foreign members are allowed to trade or purchase programs and receive monthly newsletters through the mail.

As a member of the group you will be able to purchase Spectravideo hardware and software at reduced prices from SMUG of Montreal. Non-members are permitted to attend the monthly groups meetings, by paying \$3.00 admission fee at the door.

Nous sommes le groupe officiel pour Spectravideo et MSX de la region de Montreal, tel que reconnu par Spectravideo Canada, Yamaha Canada, et Bondwell International. Les familles membres peuvent assister aux reunions du groupe, echanger des programmes et recevoir une copie du bulletin mensuel. Un membre individuel peut aussi assister aux reunions, echanger des programmes et recevoir notre bulletin. Les membres etrangers peuvent echanger ou acheter les programmes aux prix du groupe, et recevoir le bulletin par courrier.

Comme membre du groupe, vous serez autorise a acheter du materiel Spectravideo a prix reduits, chez Smug de Montreal. Les non-membres peuvent assister aux reunions de groupe en payant l'admission de \$3.00 a la porte.

MEMBERSHIP APPLICATION / APPLICATION POUR MEMBRE

Nom/Name _____ Telephone _____

Adresse/Address _____

Ville/City _____ Province _____

Pays/Cou try _____ Code Postal/Postal Code _____

Modele d'ordinateur/Computer make & model _____

Famille/Family \$30.00(CAN.)	Individuel/Individual \$20.00(CAN.)
U.S.A. \$25.00(U.S.)	Outre-Mer/Overseas \$30.00(U.S.)

Cheque Certifie ou Mandat/ Certified Cheque or Money Order - Payable to:
SMUG of MONTREAL, 28 - 159th. Ave., N.D. Ile Perrot, Que. Canada J7V 5V6

OFFICE USE ONLY - DO NOT WRITE IN SPACE BELOW.

This is to certify that the above member has paid his dues in full and is in good standing as a member of SMUG of Montreal until;

Expiration date _____ Signed and approved by _____